

AI Concepts for Official Statisticians

OS-A02-A01 Canonical Lesson

Official Statistics and AI Pathway | Controlled Candidate v0.2 | 26 September 2026

This lesson develops the technical and institutional judgement required to distinguish AI mechanisms, evaluate evidence, diagnose failure, and decide whether an AI-assisted statistical use should be permitted, redesigned, rejected or escalated. It treats AI as one component of an official-statistics evidence system and preserves the non-delegable authority of subject-matter statisticians, methodologists and designated release officials.

Control field	Value
Module	OS-A02 AI Concepts for Official Statisticians
Asset	OS-A02-A01 Canonical lesson
Academic level	Advanced master level with selected doctoral extensions
Prerequisite	OS-A01 or equivalent competence
Terminal evidence	AI Task and Technology Classification Dossier
Status	CANDIDATE FOR ACADEMIC REVIEW
Release boundary	No public release authorised

Learning outcomes and professional boundary

1. Decompose statistical work into judgement, information, transformation, prediction, generation, optimisation and control tasks.
2. Distinguish rules, statistical models, machine learning, deep learning, generative AI, retrieval systems and agents by mechanism.
3. State inputs, outputs, targets or objectives, assumptions, uncertainty and human decision boundaries.
4. Diagnose statistical, technical and human failure mechanisms and trace their propagation.
5. Connect material failures to quality, confidentiality, reproducibility, Fundamental Principles and professional authority.
6. Compare alternatives and issue a permit, redesign, reject or escalate decision supported by auditable evidence.

Outcome evidence map

Outcome	Observable evidence	Primary chapters
1 Task decomposition	Workflow map with retained judgements	3, 9, 19
2 Mechanism distinction	Defensible system classification	2, 4–9
3 Assumptions and boundaries	Inputs, objectives, uncertainty and exclusions	5–14
4 Failure propagation	Evidence graph, error pathway and controls	1, 11–17
5 Quality and authority	Quality mapping, confidentiality and stopping rules	12, 16–18
6 Comparative decision	Dossier issuing permit, redesign, reject or escalate	19–21

Professional boundary

Only public, synthetic, de-identified or formally authorised controlled information may be used. No learner may enter confidential, identifiable, embargoed or security-sensitive information into an unapproved service. AI may assist bounded tasks; it does not acquire statistical mandate, methodological authority, professional independence or release authority. Productivity does not compensate for a failed confidentiality, validity or authority gate.

Common seven-part reasoning sequence

Step	Question	Evidence
1	What statistical purpose and consequence of error?	Mandate, user, product, population, excluded use
2	What tasks and professional judgements?	Workflow map, baseline, handoffs

3	What mechanism is proposed?	Encoded, estimated, retrieved, generated or optimised function
4	What evidence would justify use?	Evaluation design, reference standard, metrics, uncertainty
5	What can fail and who is affected?	Failure register and propagation map
6	What controls and authority boundaries?	Human review, access, logging, stopping and escalation
7	What decision follows?	Permit, redesign, reject or escalate with conditions

Official statistics quality and AI assurance map

The common framework is stated once here. Later chapters apply it selectively rather than reproducing it. Quality must be evaluated at the level of the official product and its use, not only at the level of a model metric. The Fundamental Principles provide institutional obligations; the UN National Quality Assurance Framework provides a systematic quality perspective; technical AI frameworks contribute lifecycle risk practices. None replaces jurisdiction-specific law or the designated authority of the statistical institution.

Official-statistics concern	AI-related failure	Evidence and control
Relevance and public value	Optimising a convenient proxy	Defined user need, construct rationale, participation
Accuracy and reliability	Leakage, shift, biased labels, unsupported generation	Representative reference evidence, uncertainty, correction
Timeliness and cost	Fast generation with high verification burden	Quality-adjusted time and end-to-end baseline
Coherence and comparability	Model or source change breaks series	Version control, bridge studies, revision policy
Accessibility and clarity	Fluent output conceals limitations	Traceable claims, uncertainty and refusal
Confidentiality and security	Prompt, log, embedding or supplier leakage	Minimisation, access control, testing and contracts
Impartiality and independence	Automated recommendation embeds external priorities	Named authority, transparent criteria and escalation
Reproducibility	Unrecorded model, prompt, retrieval or configuration change	Evidence ledger, hashes, versions and reruns

Role division in AI-assisted statistical work

AI may support	Subject-matter statistician must retain
Search, summarisation and candidate generation	Define the statistical question and authoritative sources
Prediction, ranking and anomaly prioritisation	Define target, population, tolerances and consequences
Drafting code, tables or narrative	Verify logic, results, uncertainty and official meaning
Monitoring signals and proposed alerts	Interpret change, approve response and decide when to stop
Structured comparison of alternatives	Exercise professional judgement and record the decision

1 Official statistics as a sociotechnical evidence system

Concept and mechanism

Official statistics is not a sequence of isolated computations. It is an evidence system in which legal mandate, concepts, frames, sources, classifications, methods, infrastructure, professional judgement and public communication must remain mutually coherent. An AI component can alter several links at once: it may change which records are prioritised, which categories are proposed, how uncertainty is communicated, and who can reconstruct a decision. Technical accuracy is therefore necessary but not sufficient. A model can classify records accurately while violating the target population, reference period, disclosure rule or revision policy of the official product. (United Nations, 2014; UNSD, 2019)

Technical and statistical reasoning

Represent the system as a directed evidence graph. Nodes denote sources, transformations, estimates, decisions and releases; edges carry data, assumptions and authority. A defect at node j affects a released quantity only through reachable paths, but feedback can return the defect to later collection or training data. This representation distinguishes computational lineage from institutional authority: an executable path does not imply an authorised path. The audit unit is consequently the claim and its provenance, not merely the model file.

Implications for official statistics

The central assurance question is not whether the component appears capable, but whether its claim is supported for the intended population, production conditions and decision consequence.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for official statistics as a sociotechnical evidence system
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

1 Application analysis and advanced extension

Kintaba application

For Kintaba's proposed cross-government AI platform, the baseline is the existing controlled exchange of registers, survey files and release approvals. The proposal may reduce retrieval time, but it also introduces shared embeddings, generated summaries and opaque handoffs. Learners map every source, transformation, owner and release gate. The provisional decision is redesign: permit only bounded retrieval and comparison functions until provenance, access separation and correction loops are demonstrated.

Verification and correction protocol

Map data, methodological and authority dependencies separately. Test whether every released claim can be traced to an authorised source and named owner. Simulate a correction at an upstream node and verify that the change reaches every dependent product without bypassing revision approval.

Doctoral extension

Doctoral extension: model the evidence system as a multilayer network with data-dependency, methodological-dependency and authority edges. Ask which failures are identifiable from released outputs and which remain observationally equivalent without internal audit evidence.

Master level exercise

Draw the three-layer evidence graph and identify one failure that is visible in the release and one that requires internal audit evidence. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Graph completeness	Source, transformation, claim and authority coverage
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

2 Definitions boundaries and anthropomorphic language

Concept and mechanism

A useful definition describes observable system properties rather than attributing human faculties. The OECD definition characterises an AI system as machine-based, operating for explicit or implicit objectives, inferring from inputs how to generate predictions, content, recommendations or decisions, with varying autonomy and post-deployment adaptiveness. This definition is broad enough to include systems that learn and systems whose behaviour changes through configuration, retrieval or feedback. It does not establish competence, truthfulness, consciousness or accountability. (OECD, 2024)

Technical and statistical reasoning

Boundary analysis separates encoded rules, estimated parameters, retrieved material, generated outputs and consequential actions. Terms such as understands, decides or knows compress these distinctions and can transfer responsibility from officials to software. Replace each anthropomorphic claim with a testable statement: input domain, objective, output, adaptation mechanism, performance conditions and excluded decisions. The resulting description supports procurement, evaluation and incident review.

Implications for official statistics

The current non-AI or simpler technical method remains the comparison baseline. End-to-end quality includes verification, correction, exception handling and institutional continuity.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for definitions boundaries and anthropomorphic language
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

2 Application analysis and advanced extension

Kintaba application

A supplier claims that its assistant understands Kintaba's labour market. The institution requests a system card stating language coverage, training and retrieval sources, output types, adaptation after deployment, abstention behaviour and known exclusions. Because the supplier cannot show performance for minority-language occupation descriptions, the system may be evaluated as a candidate-code generator but not used as an autonomous coder.

Verification and correction protocol

Translate every capability claim into a falsifiable test. Evaluate language and population coverage, adaptation after deployment, refusal behaviour and excluded decisions. Reject descriptions that do not identify what is learned, encoded, retrieved or generated.

Doctoral extension

Doctoral extension: compare functional definitions used for governance with ontological claims about intelligence. Analyse how definition breadth changes the reference class for risk estimates and why regulatory classification must not be confused with scientific explanation.

Master level exercise

Rewrite five anthropomorphic supplier statements as testable system claims and specify the evidence needed for each. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Testable description	Input, objective, output, adaptation and exclusion
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

3 From statistical purpose to computational task decomposition

Concept and mechanism

Technology selection begins with the statistical purpose: the user, decision, population, concept, reference period and consequence of error. A workflow should then be decomposed into information retrieval, transformation, linkage, estimation, prediction, generation, optimisation, control and professional judgement. This prevents a vague proposal such as automate occupation coding from concealing distinct operations with different evidence requirements. (UNSD, 2019; UNECE, 2025)

Technical and statistical reasoning

A task map can be written as ordered functions T_1 through T_k with inputs, outputs, owners and acceptance conditions. Error propagation depends on composition: if $z=T_2(T_1(x))$, validation of T_2 on clean inputs does not establish performance when T_1 introduces systematic distortion. Human review is itself a task with capacity, information and error characteristics; it cannot be represented as an unlimited safeguard. Baseline measurement must record present quality, effort and failure before an AI alternative is compared.

Implications for official statistics

The subject-matter statistician specifies the construct, authoritative evidence, acceptable uncertainty and release boundary; technical staff implement and test within that mandate.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for from statistical purpose to computational task decomposition
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

3 Application analysis and advanced extension

Kintaba application

Kintaba decomposes occupation coding into capture, language normalisation, candidate generation, consistency checking, adjudication, weighting, quality measurement and release. AI is proposed only for candidate generation. Human coders retain adjudication, and methodologists retain classification and release authority. The pilot compares end-to-end corrected accuracy and review burden against the manual baseline rather than reporting classifier accuracy alone.

Verification and correction protocol

Measure the present workflow before intervention. Validate each component at its actual input boundary, then test the composed workflow. Estimate reviewer capacity and record whether automation shifts error, cost or judgement to another stage.

Doctoral extension

Doctoral extension: formulate workflow decomposition as a structural causal model. Identify where interventions on an automated component can change downstream outcomes and where selection into human review makes observed error rates non-comparable.

Master level exercise

Decompose occupation coding into at least eight tasks and identify which judgements cannot be delegated. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Workflow validity	Component and end-to-end evidence with reviewer capacity
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

4 Rules algorithms statistical models and machine learning

Concept and mechanism

Rules encode declared conditions; algorithms specify procedures; statistical models express stochastic relationships; machine-learning systems estimate functions from data. These categories overlap, but they imply different sources of change. A deterministic edit can reproduce an invalid specification perfectly. A fitted model can change when data, features, optimisation or deployment conditions change even when program code is unchanged. Governance must therefore control both specification and empirical behaviour. (UNECE, 2022)

Technical and statistical reasoning

For a rule $r(x)$, validation asks whether the encoded condition is authorised and correctly executed. For a statistical model $p(y | x, \theta)$, inference also depends on assumptions and estimation. For a learned predictor $\hat{f}$, empirical risk minimisation chooses a function that performs under a defined loss and sample distribution. None of these results alone establishes fitness for an official use: the target population, error consequences and integration into estimation must be examined.

Implications for official statistics

The central assurance question is not whether the component appears capable, but whether its claim is supported for the intended population, production conditions and decision consequence.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for rules algorithms statistical models and machine learning
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

4 Application analysis and advanced extension

Kintaba application

Kintaba compares approved census edit rules with a learned anomaly score. The score may prioritise records for review, but it cannot overwrite values or redefine an edit rule. Evaluation reports precision among reviewed cases, missed-error risk, workload concentration and regional variation. The boundary is a review queue with fixed thresholds, versioned models and periodic comparison with random quality-control samples.

Verification and correction protocol

Test rule conformance against approved specifications and learned behaviour against independent reference data. Version rules, data, features, parameters and thresholds. Compare record-level metrics with consequences for estimates and production decisions.

Doctoral extension

Doctoral extension: distinguish model-assisted production from model-assisted estimation. Analyse conditions under which prediction error affects an estimator's bias or variance and why record-level accuracy cannot substitute for design-based properties.

Master level exercise

Compare an edit rule, regression model and learned anomaly detector for the same census-control objective. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Specification and behaviour	Authorised rule plus empirical generalisation
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

5 Supervised unsupervised semi supervised and reinforcement learning

Concept and mechanism

Supervised learning estimates relationships from labelled examples; unsupervised learning seeks structure without an externally supplied target; semi-supervised learning combines limited labels with unlabelled observations; reinforcement learning adapts actions using rewards. These families differ in what constitutes evidence. Labels can encode historical practice rather than truth, clusters need substantive interpretation, pseudo-labels can amplify error, and a reward function can optimise a proxy that conflicts with public value. (Mitchell, 1997; Hastie et al., 2009)

Technical and statistical reasoning

For classification, empirical loss is an average of $L(y_i, f(x_i))$; its meaning depends on the label process, sampling, weighting and cost of different errors. Clustering stability does not prove substantive validity. Reinforcement learning additionally requires a state, action, reward and transition model; observed reward may be delayed or confounded. Official-statistics applications should generally separate exploratory structure discovery from production classification and consequential action.

Implications for official statistics

The current non-AI or simpler technical method remains the comparison baseline. End-to-end quality includes verification, correction, exception handling and institutional continuity.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for supervised unsupervised semi supervised and reinforcement learning
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

5 Application analysis and advanced extension

Kintaba application

Kintaba has 40,000 previously coded occupation descriptions and 600,000 uncoded records. In synthetic evaluation, the manual baseline achieves 91.2 percent adjudicated accuracy at 4.8 minutes per record. A supervised candidate reaches 94.6 percent before review but only 81.3 percent recall for informal-work descriptions and 76.8 percent for the two smallest language strata. The production tolerance is at least 90 percent recall in every pre-specified stratum and no increase in corrected classification error. The decision is redesign: adjudicate a stratified reference sample, repair label inconsistency and retain manual coding for low-support strata. Semi-supervised expansion is rejected until label-quality effects are measured; clustering may support taxonomy review but cannot create official categories.

Verification and correction protocol

Audit label provenance and inter-coder disagreement before training. Use stratified reference data, class-specific errors and learning curves. For clustering, test stability and substantive interpretability; for sequential action, validate the reward and safe action space.

Doctoral extension

Doctoral extension: examine learning with noisy labels and positive-unlabelled data. Derive how differential misclassification by language or sex can persist even when aggregate validation accuracy improves.

Master level exercise

Design a reference-sample strategy that exposes noisy labels and rare classes in Kintaba occupation coding. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Learning evidence	Label audit, class errors and reward validity
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

6 Representation learning and deep neural systems

Concept and mechanism

Deep neural systems learn layered representations that can reduce the need for manually designed features. Their capacity supports text, image and multimodal tasks, but it also makes performance dependent on scale, optimisation, pretraining and deployment context. A representation can preserve useful regularities while suppressing rare distinctions essential to an official classification. Apparent transferability must therefore be tested, not assumed. (Goodfellow et al., 2016)

Technical and statistical reasoning

A network composes parameterised transformations $h_l = \sigma(W_l h_{l-1} + b_l)$. Training adjusts parameters to reduce a loss, but multiple parameter settings can fit observed data, and local explanations need not reveal causal structure. Representation drift can arise when upstream encoders, tokenisers or image preprocessing change. Evaluation should include domain-specific slices, sensitivity to perturbations, reproducibility across runs and comparison with simpler baselines.

Implications for official statistics

The subject-matter statistician specifies the construct, authoritative evidence, acceptable uncertainty and release boundary; technical staff implement and test within that mandate.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for representation learning and deep neural systems
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

6 Application analysis and advanced extension

Kintaba application

Kintaba tests satellite imagery for crop-area support. In synthetic holdout data, pixel accuracy rises from 84.1 percent for the conventional classifier to 92.7 percent for the deep model, yet regional crop area is understated by 11.8 percent in intercropped zones and 15.4 percent for small fragmented plots. The tolerance for experimental publication is absolute regional relative error below 5 percent with interval coverage between 93 and 97 percent. Because the official use is area estimation rather than pixel classification, the decision is redesign: combine calibrated probability outputs with ground-survey estimators, publish domain-specific uncertainty and exclude regions lacking adequate ground truth.

Verification and correction protocol

Evaluate geographic and production-domain transfer, sensitivity to preprocessing and reproducibility across runs. Compare deep representations with simpler features. Where outputs feed estimation, quantify the effect of classification error on area totals and uncertainty.

Doctoral extension

Doctoral extension: connect representation shift to measurement invariance. Consider whether a latent representation preserves the construct across ecological zones and how representation error enters a design- or model-assisted estimator.

Master level exercise

Specify how pixel classification error enters crop-area estimation and propose a domain-specific validation design. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Transfer evidence	Domain tests and estimator-level consequences
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

7 Generative models foundation models and large language models

Concept and mechanism

Generative models estimate distributions from which new content can be produced. Large language models generate token sequences conditional on context; they do not retrieve a stored proposition each time or guarantee that fluent output is supported. Foundation models are trained broadly and adapted through prompting, retrieval, fine-tuning or tools. Their versatility creates a large claim surface: the same interface may draft, classify, translate, code or recommend, but each use requires separate evidence. (Vaswani et al., 2017; NIST, 2024)

Technical and statistical reasoning

Autoregressive generation factorises the probability of a sequence into conditional token probabilities. Decoding choices change outputs even with fixed parameters. A likelihood objective rewards plausible continuation, not factual validity, impartiality or statistical correctness. Evaluation must therefore separate task success, factual support, reproducibility, harmful failure, confidentiality and human correction cost. Prompting is a control input, not a substitute for validation.

Implications for official statistics

The central assurance question is not whether the component appears capable, but whether its claim is supported for the intended population, production conditions and decision consequence.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for generative models foundation models and large language models
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

7 Application analysis and advanced extension

Kintaba application

Kintaba pilots an LLM for drafting narrative around approved tables. Across 120 synthetic releases, the analyst baseline takes a median 52 minutes and produces 0.7 material corrections per release before sign-off. The candidate draft takes 11 minutes to generate but 35 minutes to verify and produces 1.9 material corrections, including unsupported causal language in 8.3 percent of releases. The tolerance is zero unsupported numerical claims, below 2 percent unsupported material interpretation and lower quality-adjusted time than baseline. The decision is redesign: use only public tables and controlled notes, require claim-level traceability and prohibit headline selection, unexplained interpretation and publication authority.

Verification and correction protocol

Decompose every generated output into verifiable claims. Measure material support, contradiction, omission, correction time and run-to-run variability. Test prompts as controlled inputs and include refusal cases, adversarial instructions and version changes.

Doctoral extension

Doctoral extension: distinguish epistemic uncertainty, aleatory uncertainty and generative variability. Analyse why token probability is not a calibrated probability that a claim is true, and design selective generation with abstention and verification.

Master level exercise

Design a drafting evaluation that separates factual support, interpretation quality, correction burden and variability. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Claim verification	Support, contradiction, correction and variability
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

8 Retrieval search grounding and citation architectures

Concept and mechanism

Retrieval-augmented systems separate document selection, ranking, context assembly and generation. Grounding can improve traceability only when the corpus is authoritative, current and sufficiently complete and when citations support the claims made. A polished answer with a citation can still misrepresent a source, omit a controlling document or cite a superseded version. (UNECE, 2025)

Technical and statistical reasoning

Let D be the eligible corpus, $R(q,D)$ a retrieval function and $G(q,C)$ a generator using retrieved context C . End-to-end validity requires coverage of D , recall and ranking performance of R , faithful use of C by G , and claim-level entailment. Evaluation should therefore include unanswerable questions, conflicting sources, version changes, multilingual queries and adversarially similar documents. Refusal and referral are legitimate outputs.

Implications for official statistics

The current non-AI or simpler technical method remains the comparison baseline. End-to-end quality includes verification, correction, exception handling and institutional continuity.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for retrieval search grounding and citation architectures
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

8 Application analysis and advanced extension

Kintaba application

Kintaba builds public query support from approved releases and methodological notes. In a synthetic set of 500 queries, the staff-search baseline finds the controlling source in 96.4 percent of cases; the candidate retrieves it in 93.8 percent overall but only 82.0 percent for minority-language queries. Citation support is 97.1 percent when the correct source is retrieved, while correct refusal on unsupported questions is 88.6 percent against a 98 percent tolerance. The decision is redesign: maintain source owners, validity dates and supersession links; raise multilingual recall and refusal performance; and retain referral to staff. The service may not infer confidential detail, improvise policy or treat drafts as released statistics.

Verification and correction protocol

Create a gold set covering answerable, unanswerable, conflicting, superseded and multilingual queries. Measure authoritative-source recall, ranking, claim-level citation support and correct refusal. Test corpus update and withdrawal procedures.

Doctoral extension

Doctoral extension: treat retrieval as a missing-data and selection mechanism. Analyse when conditioning generation on top-ranked documents induces systematic omission and how corpus governance changes the estimand of an answer-quality evaluation.

Master level exercise

Construct ten retrieval tests including superseded, conflicting, absent and minority-language sources. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Retrieval evidence	Corpus coverage, recall, support and refusal
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

9 Optimisation recommendation and agentic workflows

Concept and mechanism

Prediction estimates outcomes; optimisation selects actions against objectives and constraints; recommendation ranks options; an agentic workflow plans and executes multiple steps using tools, state and feedback. Moving from prediction to action changes the risk: errors can compound, actions can alter future data, and technical autonomy can be mistaken for institutional authority. (NIST, 2023; UNECE, 2025)

Technical and statistical reasoning

An optimisation problem selects a to minimise expected loss subject to constraints. The objective cannot capture every legal, ethical or statistical obligation; some conditions are hard gates rather than trade-offs. Multi-step agents additionally require tool permissions, state integrity, bounded recursion, transaction limits, logging and stopping conditions. Each step expands the audit surface, and success at individual steps does not establish safe composition.

Implications for official statistics

The subject-matter statistician specifies the construct, authoritative evidence, acceptable uncertainty and release boundary; technical staff implement and test within that mandate.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for optimisation recommendation and agentic workflows
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

9 Application analysis and advanced extension

Kintaba application

Kintaba considers an agent that schedules survey follow-up, drafts code and prepares correction files. The proposal is split. Scheduling recommendations may be piloted under capacity and equity constraints; code generation requires tests and independent review; correction files require human approval. The agent has no production credentials, cannot release outputs and stops on schema change, anomalous volume or failed validation.

Verification and correction protocol

Test each tool call independently and in composition. Enforce least privilege, transaction limits and deterministic checkpoints. Simulate tool failure, schema change, looping, stale state and an unsafe recommendation; verify containment and recovery.

Doctoral extension

Doctoral extension: examine performative prediction, where actions based on a model alter the data-generating process. Analyse equilibrium, feedback and constraint violations when recommendations change response behaviour.

Master level exercise

Define permissions and stopping conditions for an agent that schedules follow-up and drafts code without production authority. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Agent control	Permissions, checkpoints, limits and recovery
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

10 Training inference evaluation and the distinction between fit and use

Concept and mechanism

Training adjusts parameters using data and an objective; inference applies a fixed system; evaluation estimates performance under specified conditions. A development split supports tuning, while a genuinely independent test or prospective evaluation supports a stronger claim. Good fit does not establish population validity, operational feasibility or authority to use the output. (UNECE, 2022; UNECE, 2026)

Technical and statistical reasoning

Evaluation requires a declared target population, sampling mechanism, reference standard, metrics, uncertainty and decision threshold. Accuracy can conceal asymmetric costs and class imbalance. Confidence intervals should reflect the evaluation design, including clustering or weighting where relevant. External validation tests transport; prospective monitoring tests performance after integration. The baseline must be credible and evaluated under comparable conditions.

Implications for official statistics

The central assurance question is not whether the component appears capable, but whether its claim is supported for the intended population, production conditions and decision consequence.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for training inference evaluation and the distinction between fit and use
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

10 Application analysis and advanced extension

Kintaba application

For record linkage, Kintaba compares deterministic rules, probabilistic matching and a learned matcher on a synthetic stratified clerical-review sample of 8,000 candidate pairs. The probabilistic baseline has a 1.6 percent false-link rate and 7.4 percent missed-link rate; the learned matcher reduces missed links to 4.1 percent but raises false links to 2.3 percent, reaching 5.9 percent for recent migrants with unstable identifiers. Analytical estimation tolerates at most 2 percent false links after weighting, while operational register updates require below 0.5 percent plus an appeal path. The decision permits sandbox analysis with linkage-error sensitivity analysis; operational updating is rejected pending subgroup repair, prospective evidence and governance.

Verification and correction protocol

Use a temporally and operationally independent evaluation sample. Estimate false and missed decisions with design-consistent uncertainty. Compare thresholds, reviewer workload and downstream consequences against deterministic and probabilistic baselines.

Doctoral extension

Doctoral extension: distinguish conditional predictive performance from marginal population performance. Develop an evaluation estimator under unequal-probability verification sampling and examine verification bias.

Master level exercise

Estimate false-link and missed-link consequences under analytical and operational uses, then select a threshold or reject use. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Evaluation design	Independent sample, uncertainty and use-specific loss
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

11 Generalisation distribution shift and concept drift

Concept and mechanism

Generalisation is performance beyond observed training examples under a specified relationship between development and use conditions. Covariate shift changes inputs, label shift changes outcome prevalence, and concept drift changes the conditional relationship between inputs and outcomes. Official statistical systems face recurring shifts from questionnaire redesign, nonresponse, classification revision, policy change and source-system migration. (UNECE, 2022; UNECE, 2026)

Technical and statistical reasoning

A model's expected loss under production distribution Q differs from its development loss under P when distributions diverge. Importance weighting can address some covariate shift only when support overlaps and weights are stable. Drift detectors identify changes in observable signals, not necessarily harmful changes in the target relationship. Monitoring therefore combines data diagnostics, reference samples, quality indicators and institutional event logs.

Implications for official statistics

The current non-AI or simpler technical method remains the comparison baseline. End-to-end quality includes verification, correction, exception handling and institutional continuity.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for generalisation distribution shift and concept drift
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

11 Application analysis and advanced extension

Kintaba application

Kintaba's occupation coder is trained before platform-work categories expand and a new administrative portal changes text descriptions. Input drift alerts are paired with a rotating adjudicated sample. Thresholds trigger investigation, not automatic retraining. Classification experts decide whether observed change reflects terminology, population composition, concept revision or system failure.

Verification and correction protocol

Maintain event logs for questionnaire, classification, source-system and policy changes. Combine drift statistics with rotating reference samples. Test whether alarms detect harmful performance change and whether retraining improves rather than merely restores fit.

Doctoral extension

Doctoral extension: analyse transportability using selection diagrams or domain adaptation assumptions. State which invariances are required and why empirical stability over a short window cannot prove future validity.

Master level exercise

Propose a drift-monitoring plan that distinguishes terminology change from concept change and source-system failure. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Shift evidence	Event log, reference sample and response protocol
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

12 Data leakage coverage error selection bias and proxy failure

Concept and mechanism

Leakage makes unavailable or target-derived information appear in development; coverage error excludes or duplicates parts of the population; selection bias makes observed examples systematically unrepresentative; proxy failure arises when an operational measure does not adequately represent the intended construct. These are different defects and require different remedies. (UNSD, 2019; Suresh and Guttag, 2021)

Technical and statistical reasoning

Random train-test splitting does not prevent entity, time, household or geographic leakage. Performance estimation must respect the production unit and time order. Coverage and selection connect machine learning to total survey error: a precise predictor over a distorted frame can worsen an official estimate. Proxy validation requires construct reasoning, not correlation alone. Subgroup checks should be defined by consequence and data quality, not only convenience.

Implications for official statistics

The subject-matter statistician specifies the construct, authoritative evidence, acceptable uncertainty and release boundary; technical staff implement and test within that mandate.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for data leakage coverage error selection bias and proxy failure
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

12 Application analysis and advanced extension

Kintaba application

Kintaba predicts household vulnerability using service records. Rural households with weak connectivity are under-covered, future benefit information leaks into training, and service use is a proxy affected by access. The model is rejected for eligibility decisions. A redesigned analytical study separates time, audits coverage, documents the construct and reports uncertainty without assigning individual status.

Verification and correction protocol

Search explicitly for entity, household, geographic and temporal leakage. Audit frame coverage and selection into labelled data. Validate proxies against the intended construct and conduct sensitivity analysis for unobserved units and alternative definitions.

Doctoral extension

Doctoral extension: decompose total error into coverage, measurement, linkage, modelling and decision components. Analyse non-identifiability when excluded units have no observed outcomes and conduct sensitivity analysis rather than asserting correction.

Master level exercise

Build a total-error diagram for the vulnerability model and identify which errors cannot be corrected from observed data. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Total-error evidence	Leakage, coverage, selection and construct validity
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

13 Hallucination retrieval failure calibration and uncertainty

Concept and mechanism

A trustworthy workflow separates unsupported generation, retrieval failure, predictive uncertainty, sampling uncertainty, measurement uncertainty and decision uncertainty. Hallucination is not a single measurable phenomenon; operational definitions should focus on claim support, contradiction, fabrication and material consequence. Calibration asks whether stated confidence corresponds to empirical frequency under relevant conditions. (NIST, 2024; UNECE, 2025)

Technical and statistical reasoning

For probabilistic predictions, calibration can be assessed by reliability curves, proper scoring rules and subgroup analysis. A language model's expressed confidence is not automatically a probability. Retrieval systems need coverage and citation tests, while generative systems need claim decomposition and verification. Uncertainty should be retained as a ledger of sources rather than collapsed into one reassuring score.

Implications for official statistics

The central assurance question is not whether the component appears capable, but whether its claim is supported for the intended population, production conditions and decision consequence.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for hallucination retrieval failure calibration and uncertainty
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

13 Application analysis and advanced extension

Kintaba application

Kintaba's query assistant confidently cites an obsolete table. Tests include temporal traps, conflicting revisions and questions outside the corpus. The answer must display source date and release status, and it must refuse when authoritative support is absent. Monitoring records unsupported material claims, stale-source use, incorrect refusals and referral completion.

Verification and correction protocol

Maintain separate measures for predictive calibration, source retrieval, claim support and decision uncertainty. Test temporal traps and unsupported questions. Report risk conditional on answering and the distribution of abstention across user groups.

Doctoral extension

Doctoral extension: compare calibration under dataset shift and selective prediction. Evaluate coverage-risk curves when the system abstains and examine whether abstention disproportionately transfers burden to particular groups.

Master level exercise

Design a selective-answer policy and evaluate coverage, risk and unequal abstention burden. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Uncertainty ledger	Calibration, support, abstention and decision risk
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

14 Subgroup performance fairness and public value

Concept and mechanism

Fairness is not a single metric. Statistical institutions must ask who is represented, which errors matter, who bears correction costs, and whether a use is justified by mandate and public value. Equal aggregate performance can conceal failure for minority languages, remote areas, rare occupations or small enterprises. Metric selection is a normative decision informed by consequences. (UNECE, 2025; Suresh and Guttag, 2021)

Technical and statistical reasoning

False-positive parity, false-negative parity, calibration and equalised odds cannot generally all hold when prevalences differ. This mathematical incompatibility means that a dashboard cannot make the value choice. Subgroup estimates also have uncertainty; small samples require careful pooling, targeted data collection or explicit limits. Participation and appeal mechanisms complement quantitative assessment.

Implications for official statistics

The current non-AI or simpler technical method remains the comparison baseline. End-to-end quality includes verification, correction, exception handling and institutional continuity.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for subgroup performance fairness and public value
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

14 Application analysis and advanced extension

Kintaba application

Kintaba's language classifier performs well nationally but poorly for two minority-language groups. Because errors exclude records from coding assistance and increase manual burden, the pilot requires language-specific assurance, targeted reference data and guaranteed manual service. Deployment remains bounded until minimum performance and workload conditions are met.

Verification and correction protocol

Define affected groups from the intended use and error consequences. Estimate weighted subgroup errors with uncertainty, examine intersections and document unstable cells. Compare candidate fairness criteria and record the institutional value judgement.

Doctoral extension

Doctoral extension: formulate fairness under survey weights and intersectional sparse cells. Analyse partial pooling, uncertainty intervals and the risk that metric optimisation changes behaviour without improving substantive justice.

Master level exercise

Choose and defend fairness measures for minority-language coding while reporting uncertainty in sparse groups. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Distributional assurance	Weighted subgroup errors, uncertainty and participation
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

15 Automation bias deskillling compounding error and institutional memory

Concept and mechanism

Human oversight can fail when reviewers defer to confident outputs, receive too many alerts, lack time or authority, or lose manual competence. Oversight is a designed control system, not the presence of a person. Automation can also erase institutional memory when rationales, exceptions and corrections are not retained. (UNECE, 2025)

Technical and statistical reasoning

Review effectiveness depends on sampling, independence, information, workload and feedback. If reviewers inspect only low-confidence cases, high-confidence systematic errors may remain invisible. Error can compound when corrected outputs become future training data without preserving provenance. Institutions need manual fallback, rotation, blind quality samples, escalation and periodic exercises that test recovery capability.

Implications for official statistics

The subject-matter statistician specifies the construct, authoritative evidence, acceptable uncertainty and release boundary; technical staff implement and test within that mandate.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for automation bias deskillling compounding error and institutional memory
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

15 Application analysis and advanced extension

Kintaba application

Kintaba coders begin accepting high-confidence occupation suggestions without reading descriptions. The redesign inserts blinded control records, random review above the confidence threshold, reason codes for overrides and monthly error debriefs. Manual coding competence is tested annually, and model suggestions are hidden during part of the quality audit.

Verification and correction protocol

Use blinded control records, random review above confidence thresholds and override reason codes. Measure reliance, correction and reviewer fatigue over time. Test manual fallback and retained competence without model suggestions.

Doctoral extension

Doctoral extension: model joint human-machine performance rather than assuming independent errors. Examine correlated error, endogenous reliance and how interface design changes the conditional probability of human correction.

Master level exercise

Design an oversight experiment that measures whether model confidence changes human correction behaviour. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Human-system evidence	Reliance, correction, fatigue and fallback
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

16 Confidentiality cybersecurity intellectual property and supplier boundaries

Concept and mechanism

AI workflows create new data flows through prompts, embeddings, logs, model artefacts, retrieved context and supplier telemetry. Each flow can create disclosure, security, retention, jurisdiction or ownership risk. Statistical confidentiality applies across the workflow; procurement cannot transfer the institution's legal or professional responsibility. (NIST, 2023; ISO/IEC 23894, 2023)

Technical and statistical reasoning

Threat modelling should cover unauthorised access, prompt injection, model inversion, extraction, poisoning, insecure tools and supply-chain change. Data minimisation, separation, encryption, access control, retention limits and incident response must be technically enforced. Intellectual-property and licensing review should cover training data, generated code, model weights and downstream redistribution. Supplier evidence must be independently testable where consequence is material.

Implications for official statistics

The central assurance question is not whether the component appears capable, but whether its claim is supported for the intended population, production conditions and decision consequence.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for confidentiality cybersecurity intellectual property and supplier boundaries
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

16 Application analysis and advanced extension

Kintaba application

Kintaba evaluates a hosted drafting service. Confidential microdata and embargoed releases are excluded. The supplier must document data residency, training reuse, retention, subprocessors, incident notification, model-change control and deletion. Because evidence remains incomplete, only public material may be used in a segregated pilot; production procurement is escalated.

Verification and correction protocol

Trace prompts, embeddings, outputs, logs, model artefacts and supplier telemetry. Test access separation, retention and deletion. Conduct threat scenarios for injection, extraction, poisoning and supplier change; verify incident escalation.

Doctoral extension

Doctoral extension: analyse confidentiality risk under composition. Repeated low-risk interactions can create a high cumulative disclosure risk, especially when outputs, logs and external auxiliary data are linkable.

Master level exercise

Produce a data-flow and supplier-boundary assessment for the hosted drafting pilot. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Boundary evidence	Data flows, threats, contracts and incidents
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

17 Reproducibility auditability and evidence records

Concept and mechanism

Reproducibility means that controlled inputs, code, configuration, environment and versions can regenerate declared outputs within stated tolerances. Auditability is broader: an independent reviewer must reconstruct what happened, why it was allowed, who approved it and which evidence supported the decision. A nondeterministic system can still be governed if variability is measured and decision records are complete. (NIST, 2023; UNECE, 2025)

Technical and statistical reasoning

An evidence record should include source hashes, data lineage, model and prompt versions, retrieval corpus, parameters, environment, seeds where applicable, outputs, verification results, exceptions and approvals. Reproduction tests must distinguish exact equality from statistically acceptable variation. Logs must be protected, interpretable and retained according to policy rather than collected without purpose.

Implications for official statistics

The current non-AI or simpler technical method remains the comparison baseline. End-to-end quality includes verification, correction, exception handling and institutional continuity.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for reproducibility auditability and evidence records
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

17 Application analysis and advanced extension

Kintaba application

Kintaba versions the occupation model, adjudication sample, prompt, classification and thresholds. A rerun reconstructs the released quality indicators, while a decision ledger records deviations and approvals. If a supplier model changes without notice, the production gate closes until equivalence or revalidation is established.

Verification and correction protocol

Reconstruct a declared output from versioned inputs, code, environment, model, prompt and corpus. Define tolerances for stochastic components. Sample evidence records to confirm that exceptions, approvals and release decisions can be independently reproduced.

Doctoral extension

Doctoral extension: define reproducibility for stochastic pipelines using distributional tolerances and sequential tests. Consider which audit claims are falsifiable when proprietary components prevent inspection.

Master level exercise

Define exact and distributional reproducibility tests for one deterministic and one generative component. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Reproduction record	Versions, hashes, tolerances, exceptions and approvals
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

18 Human oversight professional authority and stopping rules

Concept and mechanism

Meaningful oversight assigns competent people the information, independence, time and power to intervene. A reviewer who can only approve an opaque output is not exercising control. Statistical authority remains with designated officials: systems may propose, rank or draft, but cannot acquire legal mandate, professional independence or release authority. (United Nations, 2014; UNECE, 2025)

Technical and statistical reasoning

Stopping rules translate principles into observable triggers. They should identify the monitored quantity, threshold, decision owner, immediate containment, fallback and restart conditions. Triggers may include confidentiality incidents, schema change, unsupported material claims, subgroup deterioration, unexplained volume shifts, supplier change or audit failure. Escalation is required when value choices or rights exceed the operating team's authority.

Implications for official statistics

The subject-matter statistician specifies the construct, authoritative evidence, acceptable uncertainty and release boundary; technical staff implement and test within that mandate.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for human oversight professional authority and stopping rules
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

18 Application analysis and advanced extension

Kintaba application

Kintaba defines separate rights to configure, run, review, override, suspend and release each AI-assisted task. A production operator can halt a job but cannot waive quality gates; the chief statistician can authorise release but cannot retrospectively erase failed evidence. Exercises test whether staff can stop and recover the workflow under realistic pressure.

Verification and correction protocol

Run tabletop exercises for confidentiality incident, subgroup deterioration, schema change and supplier update. Confirm who can suspend, contain, investigate, restart and release. A control fails if the designated person lacks information, time or authority.

Doctoral extension

Doctoral extension: treat oversight as a problem of institutional design and principal-agent relations. Analyse whether incentives, information asymmetry and diffusion of responsibility make nominal controls ineffective.

Master level exercise

Write operational stopping and restart rules for four material failure scenarios. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Authority test	Named powers, triggers, containment and restart
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

19 Kintaba worked cases across the statistical value chain

Concept and mechanism

Integrated cases test transfer across coding, linkage, geospatial estimation, drafting and query support. Each case contains a baseline, proposed mechanism, imperfect evidence, affected groups and a decision consequence. Competence is demonstrated by comparing alternatives and defending a bounded decision, not by approving AI whenever performance appears promising. (UNECE, 2025)

Technical and statistical reasoning

The common dossier records intended use, excluded decisions, workflow position, evaluation population, benchmark, metrics, tolerances, uncertainty, failure propagation, confidentiality, human review, stopping rules, owner and review date. Cases deliberately include evidence that points in different directions. A technically strong result can still fail a non-compensatory authority or confidentiality gate.

Implications for official statistics

The central assurance question is not whether the component appears capable, but whether its claim is supported for the intended population, production conditions and decision consequence.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for kintaba worked cases across the statistical value chain
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

19 Application analysis and advanced extension

Kintaba application

Learners compare the five cases using one decision matrix but must produce different conclusions: occupation coding may support suggestion with adjudication; linkage may remain analytical; crop-area estimation may be experimental with published limitations; release drafting may be bounded to verified drafts; query support may answer, qualify, refuse or refer. Each conclusion changes if evidence or consequence changes.

Verification and correction protocol

Require a complete dossier for each case and exchange dossiers among learners for blind review. Score consistency of mechanism classification, evidence sufficiency and decision reasoning while accepting defensible alternative conclusions.

Doctoral extension

Doctoral extension: conduct cross-case comparative analysis of error propagation and reversibility. Develop a proportional-assurance rule while preserving non-compensatory legal, confidentiality and professional-independence constraints.

Master level exercise

Issue five separate case decisions and explain why a common technology preference would be invalid. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Cross-case reasoning	Consistent fields with case-specific conclusions
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

20 Comparative technology selection and proportional assurance

Concept and mechanism

Selection should compare manual practice, deterministic rules, conventional statistical methods, machine learning, generative systems and hybrids against the same purpose and baseline. Novelty is not a benefit by itself. A simpler method may dominate when performance is similar but evidence, correction, continuity and accountability are stronger. (ISO/IEC 23894, 2023; ISO/IEC 42005, 2025)

Technical and statistical reasoning

A multi-criterion comparison can include validity, quality-adjusted time, correction burden, subgroup performance, uncertainty, confidentiality, reproducibility, reversibility, supplier dependence and institutional capability. Weighted scores may support discussion but must not compensate for mandatory gates. Assurance intensity should increase with consequence, uncertainty, irreversibility and scale.

Implications for official statistics

The current non-AI or simpler technical method remains the comparison baseline. End-to-end quality includes verification, correction, exception handling and institutional continuity.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for comparative technology selection and proportional assurance
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

20 Application analysis and advanced extension

Kintaba application

Kintaba compares manual occupation coding, rules, supervised classification and generative classification. The supervised hybrid performs best after correction, while the generative option creates more variable review effort. The decision permits a supervised suggestion pilot with manual adjudication and retains rules for stable high-precision cases. The generative option remains experimental.

Verification and correction protocol

Construct a multi-criterion comparison using quality-adjusted time, uncertainty, reversibility, continuity and institutional capability. Run sensitivity analysis on weights, then reapply mandatory gates to show which options remain inadmissible.

Doctoral extension

Doctoral extension: apply robust decision-making under deep uncertainty. Compare expected utility, minimax regret and satisficing when probabilities are weak and institutional constraints are binding.

Master level exercise

Compare four technology options under two different consequence scenarios and test sensitivity to uncertain weights. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Comparative assurance	Baseline, criteria, sensitivity and mandatory gates
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

21 Institutional decision record and workplace transfer

Concept and mechanism

A decision record converts analysis into institutional memory. It states the purpose, baseline, evidence, alternatives, conditions, accountable owners, residual uncertainty, monitoring, stopping rules and review date. It also records what was rejected and why, preventing later users from interpreting a bounded pilot as general approval. (UNSD, 2019; NIST, 2023)

Technical and statistical reasoning

Transfer requires localisation to mandate, law, language, classifications, infrastructure, workforce and data conditions. A template is not evidence. Workplace application should begin with an authorised bounded task and public, synthetic, de-identified or formally controlled information. The record must survive staff turnover and enable independent review.

Implications for official statistics

The subject-matter statistician specifies the construct, authoritative evidence, acceptable uncertainty and release boundary; technical staff implement and test within that mandate.

Question	Chapter-specific evidence
Purpose and boundary	Defined use and excluded decisions for institutional decision record and workplace transfer
Validity	Evaluation population, reference evidence, uncertainty and tolerances
Failure propagation	Path from component error to statistical product or decision
Authority	Named reviewer, stopping trigger, fallback and release owner

21 Application analysis and advanced extension

Kintaba application

Each learner completes an AI Task and Technology Classification Dossier for one Kintaba case, then prepares a 30-day transfer plan for an approved institutional task. The plan documents the current baseline, proposed mechanism, evidence plan, control plan, decision owner and review date. No exercise authorises production deployment.

Verification and correction protocol

Audit whether the record distinguishes evidence available at decision time from later outcomes. Verify owners, expiry dates, monitoring indicators and supersession. Test whether a new team can reconstruct and update the decision without informal memory.

Doctoral extension

Doctoral extension: study decision records as organisational learning infrastructure. Analyse how versioned rationales support Bayesian evidence updating without allowing later success to erase the uncertainty present at the original decision.

Master level exercise

Complete a versioned decision record and specify the evidence event that would trigger reconsideration. Submit the result as a conditional decision with one explicit uncertainty and one non-compensatory boundary.

Evidence checkpoint

Chapter evidence focus	Minimum acceptable evidence
Institutional memory	Rationale, owners, monitoring, expiry and supersession
Decision record	Baseline, result, condition, accountable owner and residual uncertainty

Integrated Kintaba comparative dossier

The terminal task compares five plausible AI-assisted uses. Learners must not select AI by default. Each decision must remain conditional on evidence and must distinguish a useful experiment from authority to produce or release official statistics.

Case	Baseline	Candidate	Primary measures	Provisional boundary
Occupation coding	Manual coding plus approved rules	Supervised or generative candidate codes	Corrected accuracy, rare-class recall, review time, drift	Suggestion with adjudication
Record linkage	Deterministic and probabilistic linkage	Learned matching	False and missed links by use and subgroup	Analytical sandbox before operations
Crop area	Ground survey and conventional estimation	Remote-sensing classification plus estimation	Area error, coverage, calibration, regional uncertainty	Experimental estimate with limitations
Release drafting	Analyst-authored narrative	Language-model draft	Material claim support, correction burden, reproducibility	Draft only; no autonomous release
Public query support	Search and staff referral	Retrieval-augmented response	Authoritative recall, citation support, refusal and referral	Answer, qualify, refuse or refer

Required dossier

1. State purpose, user, population, consequence and excluded decisions.
2. Map workflow tasks, professional judgements and current baseline.
3. Classify the candidate mechanism and all material dependencies.
4. Specify evaluation design, benchmarks, tolerances, subgroups and uncertainty.
5. Register failure mechanisms and propagation into products or decisions.
6. Define confidentiality, verification, correction, human review, stopping and escalation.
7. Issue and defend a conditional decision with monitoring and review date.

Synthesis and thirty day workplace transfer

AI literacy for official statisticians is the ability to connect computational mechanisms to the evidentiary and institutional conditions under which official statistics earns trust. It is demonstrated by explicit task decomposition, credible comparison, verification, uncertainty, professional authority and willingness to stop. Prompt fluency and familiarity with a vendor interface are insufficient.

Field	Learner entry
Approved bounded task	Describe one task without protected information
Current baseline	Document method, effort, quality and known failures
Candidate mechanism	State what is encoded, learned, retrieved, generated or optimised
Evidence plan	Define population, reference standard, metrics, subgroups and uncertainty
Control plan	Define access, verification, correction, logging, stopping and fallback
Authority and date	Name reviewer, decision owner, release owner and review date

Final professional reflection

A competent practitioner can explain what a system does, what its evaluation establishes, what remains unknown, how errors propagate, who bears consequences, which controls are effective and who has authority to proceed. The defensible outcome may be permission for a bounded pilot, a redesign, a rejection or escalation. Each is a positive professional result when supported by evidence.

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Reference control

Normative authority, official-statistics guidance, technical standards and scholarly explanation serve different purposes. Jurisdiction-specific law, classifications, security requirements and institutional mandates must be localised before delivery. References and framework versions must be checked at scheduled review and after material change.